

Cable-Driven Manipulator + Exosuit

Progress: Simulation, Control Stack, and Rehab-Inspired Adaptation

Dipankar Bhattacharya
with Dagmar Sternad & Salah Bazzi

May 20, 2026

Dyson School of Design Engineering | Morph Lab, Imperial College London

Simulation Platform

System Architecture (Top-Down)

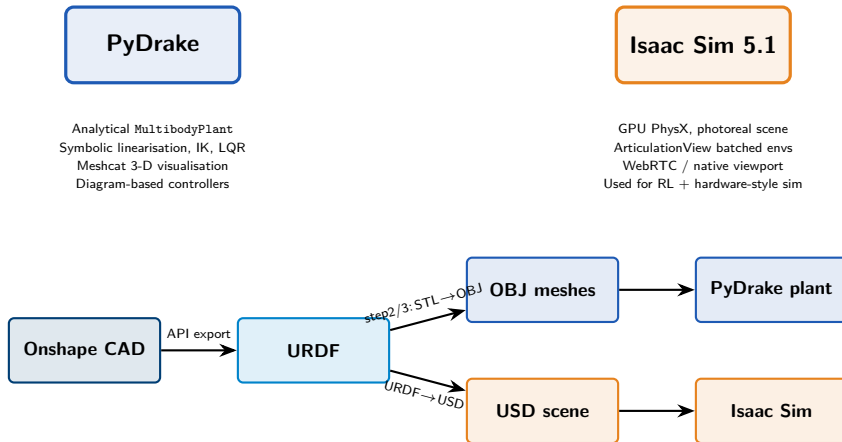
Behaviour: Transparent vs Activated

Rehab-Inspired Parameter Learning

Wrap-up

Simulation Platform

Two complementary simulators



Single source of truth: the Onshape assembly drives both pipelines.

`model_using_onshape_to_robot/manipulator_cable_exo_springs/`

Pipeline (scripted):

1. `step1_convert_from_onshape_*.sh`
Onshape API $\rightarrow$ URDF + STL assets
2. `step2_convert_stl_to_obj.py`
STL $\rightarrow$ OBJ (Drake/Isaac friendly)
3. `step3_urdf_stl_to_obj.py`
Rewrite URDF mesh references
4. `step4a_urdf_fix_inertia_for_exo_manip.py`
Inertia regularisation



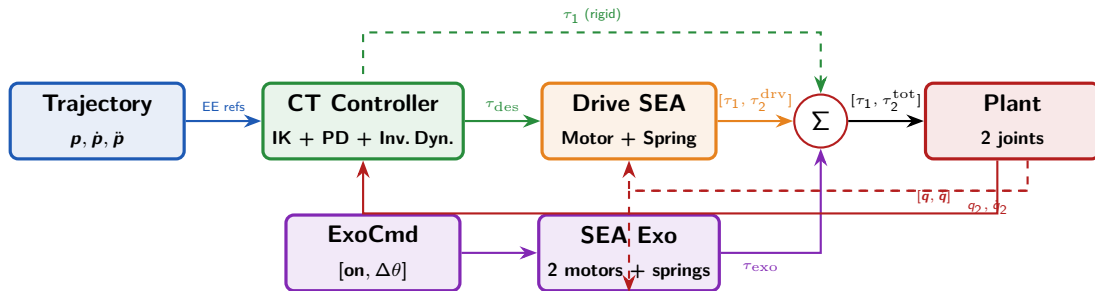
Cable manipulator + exo (Drake render)

What we get

- Identical kinematics in both simulators
- Pulley + idler frames carried over
- Reproducible: rebuild from CAD any time

System Architecture (Top-Down)

Three-layer control stack — the whole picture



Reading the diagram: the CT controller commands joint torques; joint 2 must pass through a series-elastic cable (Drive SEA) before reaching the plant. The Exo is a second SEA pair acting in parallel on joint 2.

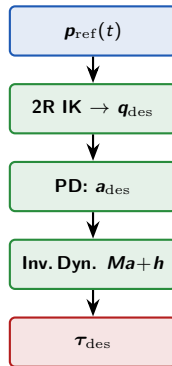
Layer 1 — Computed-Torque controller

$$\tau_{\text{des}} = \mathbf{M}(\mathbf{q}) \mathbf{a}_{\text{des}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}})$$

$$\mathbf{a}_{\text{des}} = K_p(\mathbf{q}_{\text{des}} - \mathbf{q}) - K_d \dot{\mathbf{q}}$$

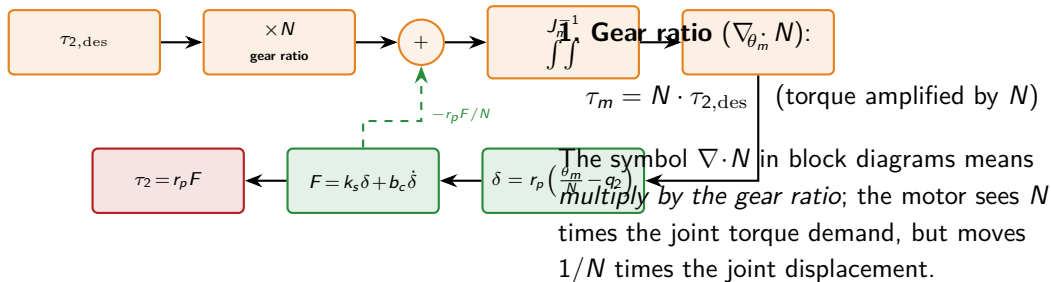
- $\mathbf{q}_{\text{des}}$ from numerical IK on $\mathbf{p}_{\text{ref}}(t)$
- Feedback linearises rigid-body dynamics
- Saturates at motor peak torque

Joint 1 is direct/rigid; only joint 2 has a cable.



Layer 2 — Drive SEA (joint 2 only)

Key equations



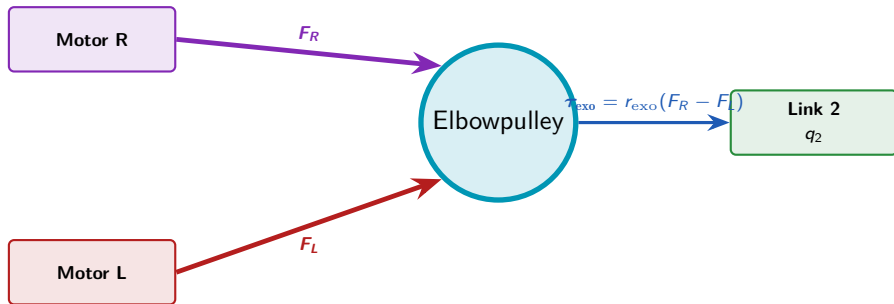
2. Rotor dynamics:

$$J_m \ddot{\theta}_m = \tau_m - b_m \dot{\theta}_m - \frac{r_p F}{N}$$

3. Spring extension & force:

$$\delta = r_p \left(\frac{\theta_m}{N} - q_2 \right), \quad F = k_s \delta + b_c \dot{\delta}$$

Layer 3 — Exosuit (Method B: centred elbow pulley)



Motor commands — co-contraction pattern:

$$\frac{\theta_{mR}}{N} = q_{2,\text{des}} + \Delta\theta, \quad \frac{\theta_{mL}}{N} = -q_{2,\text{des}} + \Delta\theta$$

where $\Delta\theta$ *pre-tensions both cables equally*.

Cable force (each, one-sided):

Net exosuit torque:

$$\tau_{\text{exo}} = r_{\text{exo}}(F_R - F_L)$$

Linearised impedance (around $q_{2,\text{des}}$):

$$\tau_{\text{exo}} \approx -K_{\text{eff}}(q_2 - q_{2,\text{des}}) - B_{\text{eff}}\dot{q}_2$$

Behaviour: Transparent vs Activated

Two operating modes of the exo

Transparent (off)

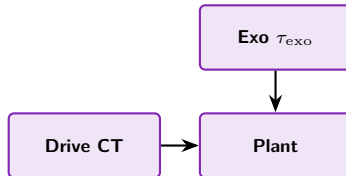
- Each motor tracks the *actual* q_2
- Springs stay at rest: $\delta_{R/L}=0$
- $\tau_{\text{exo}} \approx 0$ ✓ the user "doesn't feel" the suit
- Baseline tracking performance



exo transparent
 $\tau_{\text{exo}} = 0$

Activated (co-contraction)

- Both cables pre-tensioned: $\Delta\theta > 0$
- Springs deflect with motion
- Adds joint stiffness K_e + damping B_e
- Resists deviations *about* $q_{2,\text{des}}$



exo active — adds K_e , B_e at joint 2

Disturbance test — circle trajectory + sinusoidal kick

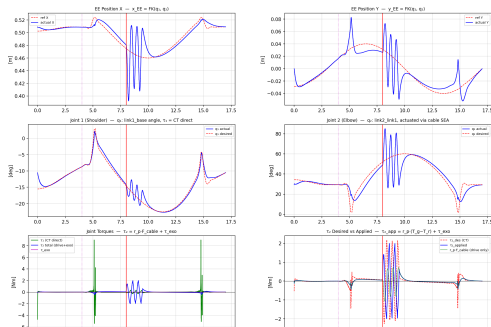
Scenario. The EE tracks a circle. At $t = 8$ s a 2 Hz, 2 Nm sinusoidal torque is injected on joint 2 for ~ 3 cycles. We compare:

- ❌ **Exo OFF** (transparent): drive CT must handle everything
- ✅ **Exo ON** (dist-sync): exo activates exactly during the disturbance

Exo OFF

SEA + Exo Co-Contraction — $k_{ss}=200.0$ $k_{exo}=8000.0$ $\delta\theta=0.105$ $k_{eff}=36.4810$ Nm/rad | EXO: OFF (deactivated)
(Manipulator & Drive SEA)

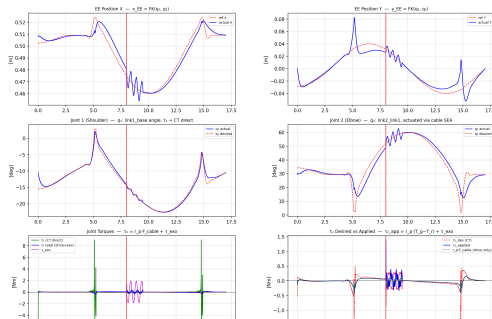
EXO: OFF (deactivated)



Exo ON (dist-sync)

SEA + Exo Co-Contraction — $k_{ss}=200.0$ $k_{exo}=8000.0$ $\delta\theta=0.105$ $k_{eff}=36.4810$ Nm/rad | EXO: DIST-SYNC (t=[8.0-9.5 s])
(Manipulator & Drive SEA)

EXO: DIST-SYNC (t=[8.0-9.5 s])

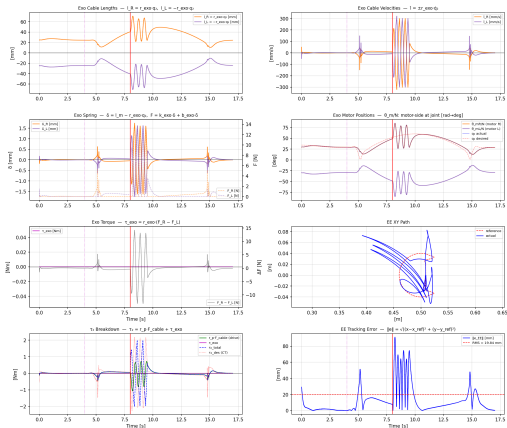


Exo-side diagnostics — where the help comes from

Exo OFF

SEA + Exo Co-Contraction — $k_{s,co}=200.0$ $k_{s,exo}=8000.0$ $\delta\theta=0.100$ $k_{s,eff}=36.4810$ Nm/rad | EXO: OFF (deactivated)
(Exosuit Co-Contraction Detail)

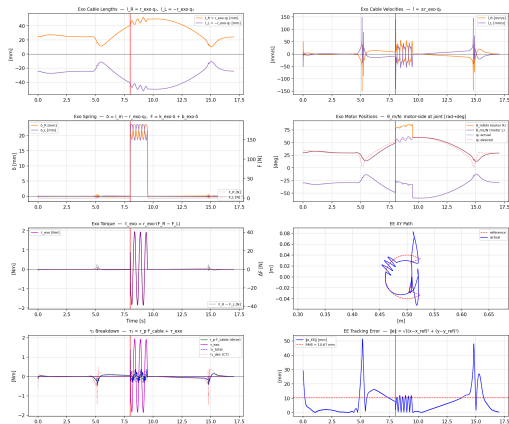
EXO: OFF (deactivated)



Exo ON (dist-sync)

SEA + Exo Co-Contraction — $k_{s,co}=200.0$ $k_{s,exo}=8000.0$ $\delta\theta=0.100$ $k_{s,eff}=36.4810$ Nm/rad | EXO: DIST-SYNC (t=8.0-9.5 s)
(Exosuit Co-Contraction Detail)

EXO: DIST-SYNC (t=8.0-9.5 s)



Key observations

Rehab-Inspired Parameter Learning

Why learn the exo impedance?

Clinical analogy

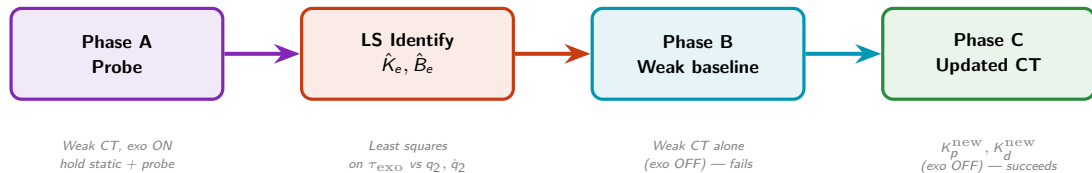
- Patient (= weak CT) tries to move
- Exo (= therapist hand) adds stiffness around the desired trajectory
- Patient *learns* how much help they were receiving
- Therapist gradually withdraws, patient succeeds alone

Engineering picture

- Weak gains $K_p^{\text{weak}}, K_d^{\text{weak}}$ cannot track alone
- Exo provides K_e, B_e during a probing phase
- We identify $\hat{K}_e, \hat{B}_e$ from data
- Update CT gains, switch the exo *off*
- Now CT alone tracks well

The exo is a *teacher*, not a permanent crutch.

Three-phase workflow



Phase A: probe. Inject $\tau_p(t) = A_p \sin(\omega_p t)$ at joint 2; exo is active with offset $\Delta\theta = 0.1$ rad. Log $q_2, \dot{q}_2, \tau_{\text{exo}}$.

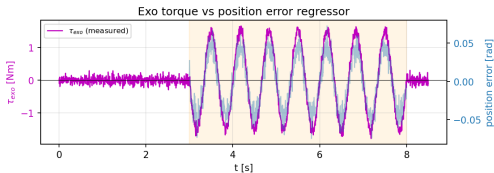
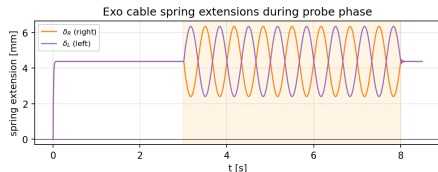
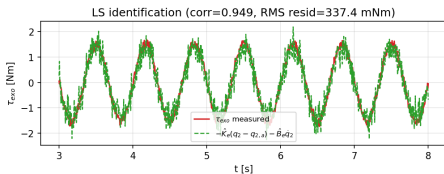
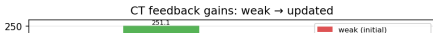
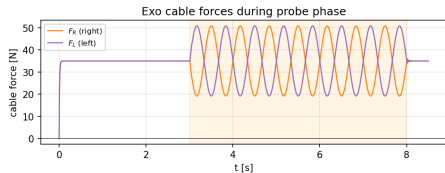
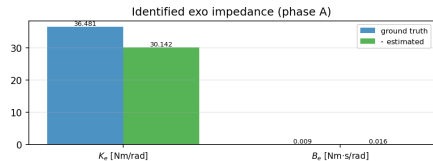
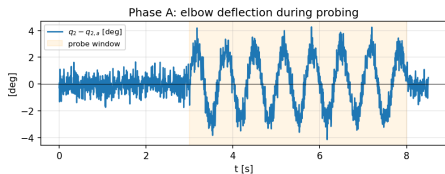
Identification. Stack the regressor $\Phi = [-(q_2 - q_{2,a}), -\dot{q}_2]$ and solve $[\hat{K}_e, \hat{B}_e]^\top = (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{y}$ where $\mathbf{y} = \tau_{\text{exo}}$.

Phase C: gain update (additive).

$$K_p^{\text{new}} = K_p^{\text{weak}} + \alpha_K \hat{K}_e, \quad K_d^{\text{new}} = K_d^{\text{weak}} + \alpha_D \hat{B}_e$$

Learning result — full diagnostic overview

Probe-based exo impedance learning → CT gain adaptation
Ground-truth: $K_e=36.481$ Nm/rad, $B_e=0.0091$ Nm·s/rad | Estimated: $K_e=30.142$ Nm/rad, $B_e=0.01563$ Nm·s/rad



Phase B/C: End-effector trajectory

Learning result — panel-by-panel explanation

Phase A — Probing (left column)

- **Top:** $q_2 - q_{2,\text{des}}$ oscillates during the probe window (orange shading) as a 1.5 Hz sine torque is injected. The deflection is rich enough for identification.
- **Bar chart:** $\hat{K}_e = 33.4 \text{ Nm/rad}$ vs. ground truth 36.5 (**only 8% error**). $\hat{B}_e$ is biased by unmeasured joint friction.
- **Cable forces:** F_R, F_L oscillate in anti-phase around the pre-tension baseline, confirming both antagonistic cables are active.
- **CT gains updated:** $K_p : 10 \rightarrow 247$; $K_d : 2 \rightarrow 4.4$. Drive feedback is now strong enough to track without the exo.
- **Tracking error:** updated CT (green)

Phase ID — LS fit (right column)

- **LS fit:** measured τ_{exo} (solid) vs. fitted $-\hat{K}_e(q_2 - q_{2,a}) - \hat{B}_e\dot{q}_2$ (dashed). Correlation = 1.000, RMS residual = 0.2 mNm — fit is essentially perfect.
- **Spring extensions:** right cable extends while left compresses — the antagonistic spring behaviour of Method B.
- **Regressor quality:** $-(q_2 - q_{2,a})$ and $-\dot{q}_2$ are phase-shifted sinusoids — linearly independent, so the LS matrix $\Phi^\top \Phi$ is well-conditioned.

Phase B/C trajectory (bottom right)

Red = weak CT alone drifts off-circle; green = updated CT (exo OFF) follows the circle

Script 14 Series: Controller Progression

Scripts 14a–14j: Controller Progression on Cup Manipulator + SEA + Exo

Script	Controller	Key Feature
14a	Computed Torque (CT)	MATLAB port of PyDrake reference; baseline tracking
14b	CCM constant metric ($M=P$)	Swaps CT for LQR-Riccati CCM; exo improves λ automatically
14c	CCM + regional exo	Gaussian mixture of K_{eff} over joint-space regions
14d	CT + RL residual	Scaffold: $\tau = \tau_{\text{CT}} + \tau_{\text{RL}}$; 2-layer MLP via REINFORCE
14e	CCM SDP, <i>no exo</i>	YALMIP block-LMI bisection; finds λ_{max} baseline
14f	CCM SDP, <i>with exo</i>	Exo adds $-K_{\text{eff}}$ term $\Rightarrow \lambda_{\text{max}}(\text{exo}) > \lambda_{\text{max}}(\text{no-exo})$
14g	CCM Slotine formulation	Original SDP (not block-LMI); no-exo vs with-exo side-by-side
14h	C3M neural controller	Learned $w_1(x), w_2(x)$ gain matrices; 6D state, exo switched
14i	C3M + embedded linear exo	Exo always in $f(x)$; fixes train/deploy mismatch of 14h
14j	C3M, 10D augmented state	Adds 4 exo motor states; full certified dual-motor control

Progression logic

- **14a:** reference baseline (CT)
- **14b–c:** CCM tracking, uniform then regional

Key takeaways

- Passive exo *provably* increases contraction rate λ (14e vs 14f)
- CCM $\approx$ CT near origin; advantage grows

Wrap-up

Done

- Onshape → URDF → OBJ & USD pipeline
- CT + Drive SEA + Exo (Method B) in PyDrake
- Matching Isaac Sim scene
- Disturbance suite:
vel/pos/torque/sine/pulse/square
- Probe-based identification of K_e, B_e
- Additive CT gain update, exo hand-off works in sim

Next

- Bias-free damping identification (separate friction)
- Reactive exo activation policy (error-triggered)
- Contraction-theory based stability proof
- Sim-to-real bridge: motor command logs → AK60-6 hardware
- Subject-study analogue: vary $\Delta\theta$, learning rate

Thanks!

Happy to dive into any block.